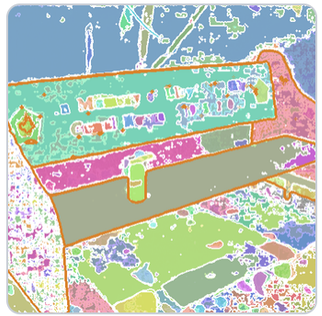
(a) Ask, don't draw: masks become tokens, the MLLM points at one

Query *"Locate the seat of the bench."*



SAM 3 (frozen) → grid + text



259 candidate masks

point grid SAM 3 text

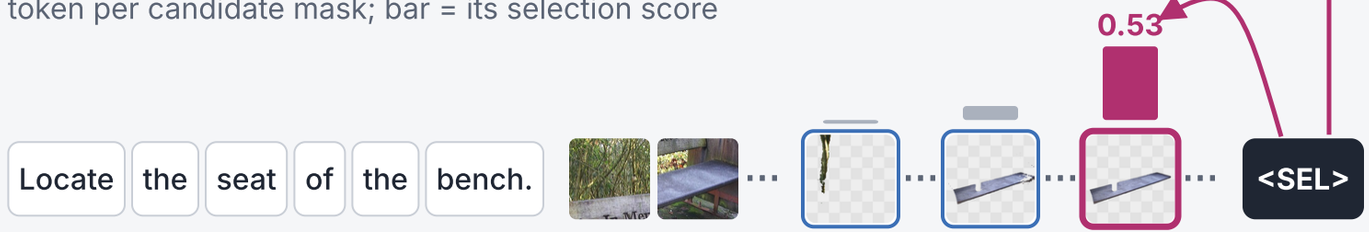
MLLM selects (no mask decoder)



selected mask

MLLM input sequence Qwen3-VL-4B, 64.2M trainable parameters

one token per candidate mask; bar = its selection score



(b) One read-out, three kinds of answer

Part

"Locate the fork of the bicycle."



1 mask, IoU 0.68

PACO +14.5

Several targets

"right man and the cake"



2 masks, IoU 0.95

gRefCOCO +20.8

No target

"man on left"



max σ 0.86 < 0.90

N-acc 89.6%